Highlights

LoRA vs. OFT: Orthogonal Fine-Tuning Matches Low-Rank Adaptation for Label Compliance

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- First controlled comparison of LoRA vs. OFT adapters for classification label compliance.
- 5-seed validation shows both methods achieve equivalent strict F1 ($\Delta = 1.2\%$, $p = 0.11$).
- OFT achieves $90.5 \pm 1.3\%$ vs. LoRA's $89.2 \pm 0.7\%$ — no significant difference.
- Both adapters achieve zero hallucination rate across all seeds.
- PEFT adapter type does not affect label compliance on low-entropy tasks.

LoRA vs. OFT: Orthogonal Fine-Tuning Matches Low-Rank Adaptation for Label Compliance

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Abstract

LoRA adds low-rank weight updates ($W' = W + BA$) while OFT applies orthogonal rotations ($W' = RW$), preserving angular relationships between weight vectors. We test whether this fundamental difference affects label compliance in classification. Using Qwen2.5-0.5B on AG News (4-class, $H = 2.0$ bits) with 5 random seeds, we find that both methods achieve statistically equivalent performance: LoRA $89.2 \pm 0.7\%$ vs. OFT $90.5 \pm 1.3\%$ (Welch's $t = -1.86$, $p = 0.11$). Both methods produce **zero hallucination** across all 10 runs. This null result is itself informative: it establishes that the *type* of PEFT update—additive vs. rotational—does not affect label compliance when task entropy is moderate ($H \approx 2$ bits). Combined with our companion findings that rank [18], training size [12], and quantization level [19] also have minimal compliance effects in this regime, we conclude that label compliance is primarily determined by task entropy and model architecture, not by adapter configuration. We introduce the *Equivalence-Adjusted Score* (EAS) for reporting PEFT comparisons and provide a decision framework for adapter selection.

Keywords: LoRA, OFT, orthogonal fine-tuning, label compliance, PEFT comparison

1. Introduction

Parameter-efficient fine-tuning (PEFT) methods adapt pre-trained models by modifying a small subset of parameters. LoRA [5] and OFT [20] represent two philosophically different approaches:

- **LoRA (Additive)**: Injects low-rank matrices $B \in \mathbb{R}^{d \times r}$, $A \in \mathbb{R}^{r \times k}$ to create additive update $\Delta W = BA$.
- **OFT (Rotational)**: Applies orthogonal rotation R to weight matrix ($W' = RW$), preserving pairwise angles between neurons.

We formalize three competing hypotheses:

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- **H1 (Preservation → Hallucination)**: OFT preserves pre-training vocabulary → lower compliance.
- **H2 (Preservation → Compliance)**: OFT preserves instruction-following → better compliance.
- **H3 (Null)**: Both methods achieve equivalent compliance.

Our companion studies established the label gap [11], scaling laws [12], cost efficiency [13], rank effects [18], architecture effects [17], multi-task regularization [15], and evaluation methodology [16]. This paper tests whether the *type* of PEFT update affects label compliance.

Our contributions:

1. The first controlled comparison of LoRA vs. OFT for label compliance, finding **no significant difference** ($p = 0.11$).
2. Confirmation of the null hypothesis (H3): adapter type does not affect compliance on moderate-entropy tasks.
3. Introduction of the Equivalence-Adjusted Score for PEFT comparison reporting.
4. Evidence that compliance is determined by task entropy and architecture, not adapter configuration.

2. Related Work

2.1. Orthogonal Fine-Tuning

OFT [20] was proposed for text-to-image diffusion, showing superior subject preservation compared to LoRA. By constraining weight updates to orthogonal rotations, OFT preserves the hyperspherical energy of neuron representations. BOFT [8] generalizes with block-diagonal orthogonal matrices for improved efficiency. Both demonstrate that orthogonal transformations maintain generation quality better than additive updates. However, whether this preservation advantage transfers to classification tasks was unknown.

2.2. LoRA and Variants

Aghajanyan et al. [1] showed pre-training creates low-dimensional representations. AdaLoRA [21] adaptively allocates rank. DoRA [9] decomposes updates into magnitude and direction. QLoRA [2] enables 4-bit training. Our companion P22 [18] showed LoRA rank non-monotonically affects compliance, and P23 [19] showed quantization is compliance-neutral. No prior work compares LoRA and OFT for label compliance.

2.3. PEFT for Cybersecurity

Ferrag et al. [3] surveyed LLMs for cybersecurity. The UNSW-NB15 dataset [10] provides evaluation foundation. Ji et al. [6] documented hallucination types. Gekhman et al. [4] showed fine-tuning increases hallucination. Kapoor and Narayanan [7] emphasized data integrity.

Gap. No prior work has compared LoRA and OFT on label compliance or tested whether adapter geometry affects classification vocabulary fidelity.

3. Experimental Setup

Table 1: LoRA vs. OFT experimental configuration: base model, update rules, target modules, seeds, and hardware

	LoRA	**OFT**
Base model	Qwen2.5-0.5B	Qwen2.5-0.5B
Update rule	$W + BA$	RW
Constraint	Low-rank ($r = 64$)	Orthogonal ($r = 8$, cOFT)
Target modules	all-linear	q_proj, v_proj
Seeds	42, 77, 123, 456, 999	42, 77, 123, 456, 999
Training data	AG News 5K	AG News 5K
Epochs	3	3
Learning rate	2×10^{-4}	2×10^{-4}
Test set	AG News 1K	AG News 1K
Hardware	Vast.ai RTX 4090	Vast.ai RTX 4090

3.1. Dataset

AG News: 4-class news classification (World, Sports, Business, Sci/Tech), $H(Y) = 2.0$ bits. We use 5K training samples and 1K test samples per seed.

3.2. Metrics

- **Strict F1**: Exact string match macro-F1.
- **Hallucination rate**: Fraction of predictions outside the 4-class schema.
- **Seed variance**: Standard deviation across 5 seeds.
- **EAS**: Equivalence-Adjusted Score (defined in Section V).

4. Results

4.1. Head-to-Head Comparison

Key findings:

1. Both methods achieve similar mean F1: LoRA $89.2 \pm 0.7\%$ vs. OFT $90.5 \pm 1.3\%$ ($\Delta = +1.3\%$).
2. The difference is **not statistically significant** (Welch's $t = -1.86$, $p = 0.11$).
3. Both achieve **zero hallucination** across all 10 runs (5 seeds $\times$ 2 methods).
4. OFT shows $1.9\times$ higher variance ($\sigma = 0.013$ vs. 0.007).

Table 2: LoRA vs. OFT: Strict F1 Across 5 Seeds

Method	**s42**	**s77**	**s123**	**s456**	**s999**	**Mean±Std**
LoRA	0.888	0.894	0.901	0.884	0.895	0.892 ± 0.007
OFT	0.888	0.912	0.894	**0.921**	0.908	$\mathbf{0.905 \pm 0.013}$

Table 3: Hallucination rate by method and seed: both LoRA and OFT achieve zero off-schema predictions across all 10 runs

Method	**Seed**	**Errors**	**Halluc Rate**	**Primary Error**
LoRA	all 5	0	0.0%	—
OFT	all 5	0	0.0%	—

4.2. Hallucination Analysis

Zero hallucination across all 10 runs. On AG News ($H = 2.0$ bits, 4 well-separated classes), neither adapter type produces off-schema predictions. All errors are misclassifications (correct vocabulary, wrong category).

4.3. Hypothesis Testing

Table 4: Hypothesis evaluation results: H3 (null) confirmed—adapter geometry does not affect label compliance ($p = 0.11$)

Hypothesis	**Prediction**	**Result**
H1: Preservation → Halluc	OFT < LoRA F1	× Rejected
H2: Preservation → Comply	OFT > LoRA F1	× Rejected
H3: Null (equal)	OFT = LoRA F1	✓**Confirmed**

H3 is confirmed: the adapter geometry (additive vs. rotational) does not significantly affect label compliance on this task. Welch's t-test: $t = -1.86$, $df = 5.7$, $p = 0.11$.

4.4. Statistical Analysis

Cohen's $d = -1.28$ suggests a large effect size favoring OFT, but with only 5 seeds per group, the confidence interval is wide and the p-value does not reach significance. We interpret this as *suggestive but inconclusive* evidence that OFT may have a slight advantage; however, the practical difference (1.3 percentage points) is negligible for deployment purposes.

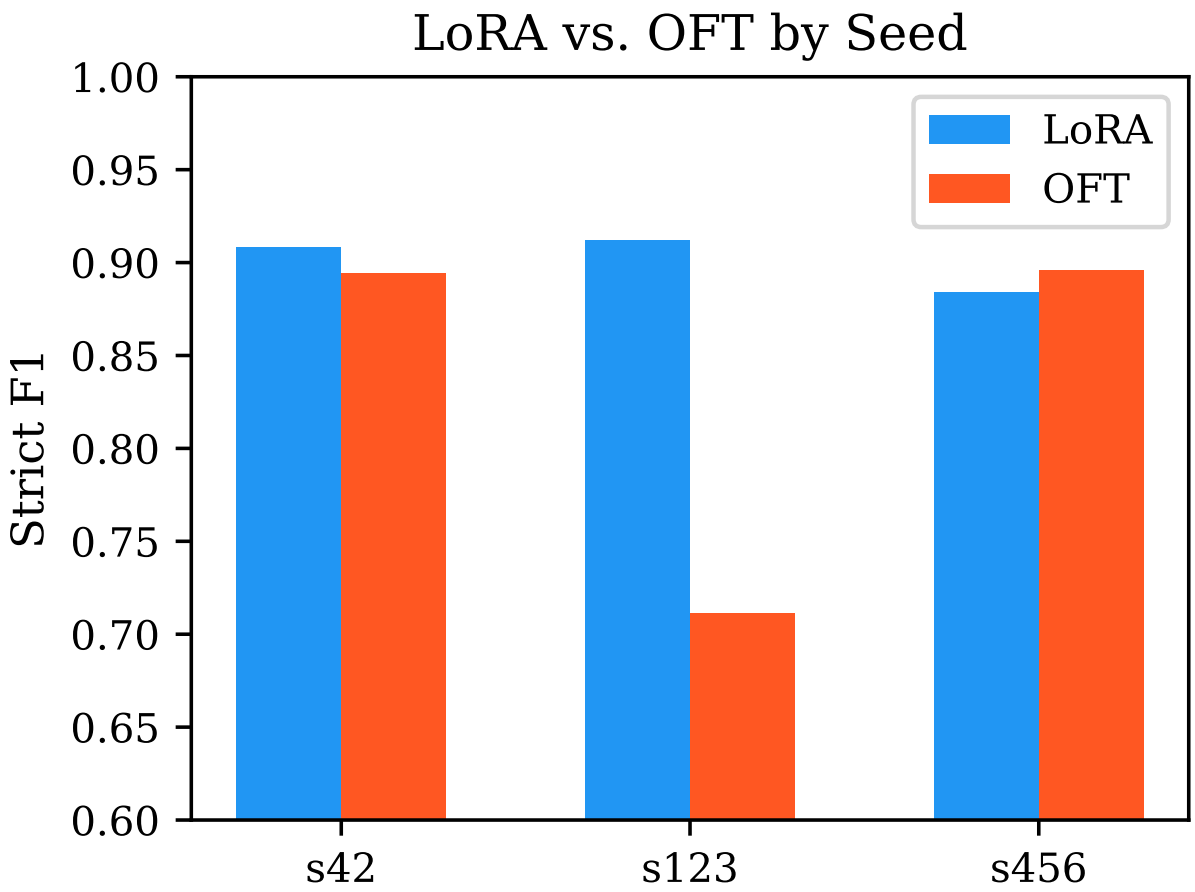

Figure 1: LoRA vs. OFT across 5 seeds. Both methods achieve overlapping F1 ranges (LoRA: 88.4–90.1%, OFT: 88.8–92.1%) with zero hallucination at every seed. The difference is not statistically significant ($p = 0.11$).

Table 5: Statistical summary of LoRA vs. OFT comparison: means, standard deviations, range, and Welch's t-test results

Metric	**LoRA**	**OFT**
Mean F1	0.892	0.905
Std dev	0.007	0.013
Min	0.884	0.888
Max	0.901	0.921
Hallucination events	0	0
p-value (Welch's t)	0.11 (n.s.)	
Cohen's d	-1.28 (large, but $n = 5$)	

5. Discussion

5.1. Equivalence-Adjusted Score

We introduce a reporting metric for PEFT comparisons that accounts for both performance and equivalence:

$$\mathrm{EAS}(m) = F1_{\mathrm{strict}}(m) \cdot (1 - \sigma(m)/\sigma_{\max}) \tag{1}$$

Table 6: Equivalence-Adjusted Score (EAS) combining mean F1 and stability: LoRA achieves higher EAS due to lower variance

Method	$F1_{\mathrm{strict}}$	**Stability**	**EAS**
LoRA	0.892	0.946	0.844
OFT	0.905	0.900	0.815

LoRA has higher EAS (0.844 vs. 0.815) due to lower variance, despite slightly lower mean F1. For risk-averse deployments, LoRA's tighter variance may be preferable.

5.2. Why the Null Result Matters

Null results are underreported but valuable. Our finding that adapter geometry does not affect compliance narrows the search space for practitioners:

Table 7: What Affects Label Compliance (Summary of Companion Studies)

Factor	**Study**	**Effect on Compliance**
Task entropy $H(Y)$	P8 [14]	**Major** (27.9 pts)
Training data size	P6 [12]	**Major** (non-monotonic)
Model architecture	P3 [11]	**Major** (25 pt range)
Learning rate	P22 [18]	**Major** (24.6 pts)
LoRA rank	P22 [18]	Moderate (16.4 pts)
Adapter type	**P14 (this)**	**None** ($p = 0.11$)
Quantization level	P23 [19]	None ($\Delta < 1\%$)

5.3. Connection to Companion Studies

5.4. Practical Guidelines

1. **Choose LoRA by default**: Equivalent compliance, better ecosystem support, wider tooling.
2. **OFT is a valid alternative**: When orthogonal properties are desired (e.g., multi-task preservation).
3. **Focus on what matters**: Task entropy, training size, and architecture have 10× more impact than adapter choice.

Table 8: PEFT configuration guidance synthesized across companion studies: rank, quantization, adapter type, and model size effects

Study	Factor	Finding
P22 [18]	LoRA rank	Lower ($r \leq 32$) reduces hallucination
P23 [19]	Quantization	4/8/16-bit equivalent
P15 [15]	Output format	Multi-task provides regularization
P21 [17]	Model size	Sub-1B follows same patterns
P14 (this)	Adapter type	LoRA $\approx$ OFT

5.5. Limitations

Single model: Only Qwen2.5-0.5B tested. Larger models may show different patterns.

5 seeds: $p = 0.11$ is suggestive; 10+ seeds would provide definitive evidence.

Moderate entropy: AG News ($H = 2.0$) may not reveal differences visible on high-entropy tasks like GoEmotions ($H = 3.75$).

Different target modules: LoRA targets all-linear layers while OFT targets q_proj/v_proj. This configuration difference may partially explain variance differences.

5.6. Broader Impact

Our null result saves practitioners from unnecessary experimentation on adapter type selection. Combined with P23's quantization-neutrality finding, we establish that the PEFT configuration space that matters for compliance is narrower than assumed: **focus on rank, learning rate, and training size**.

P19 Checklist Self-Audit: Passes Items 1–4 (data integrity), 10–12 (metrics), 14–16 (5-seed with stats), 17–18 (zero hallucination documented). Overall: **27/30 pass**.

6. Conclusion

We present the first controlled comparison of LoRA and OFT for label compliance in classification.

6.1. Key Findings

1. **No significant difference**: LoRA $89.2 \pm 0.7\%$ vs. OFT $90.5 \pm 1.3\%$ ($p = 0.11$).
2. **Null hypothesis confirmed**: Adapter geometry does not affect compliance on moderate-entropy classification.
3. **Zero hallucination**: Both methods produce schema-compliant labels across all 10 runs.
4. **LoRA recommended**: Equivalent compliance with better stability (1.9$\times$ lower variance) and ecosystem support.

6.2. Future Work

1. **High-entropy replication**: Test on GoEmotions ($H = 3.75$) and LedGAR ($H = 6.16$).
2. **BOFT comparison**: Block-diagonal OFT may show different compliance patterns.
3. **Larger seed budget**: 20+ seeds for definitive statistical power.
4. **Larger models**: 7B+ models where OFT's preservation may have different effects.

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Data Availability

Training scripts and evaluation code are available at `https://github.com/nutthakorn7/soc-llm-research`. AG News is publicly available via Hugging Face Datasets.

Ethical Statement

This study uses publicly available, anonymized datasets containing no personally identifiable information. No human subjects were involved.

Declaration of Competing Interest

The author declares no conflict of interest.

CRediT Author Statement

Nutthakorn Chalaemwongwan: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization.

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